HAlg Exam Sheet 2025/26 - What Doesn't Kill You

Definition: Random Definitions

- $\mathrm{pr}_i : X_1 \times \ldots \times X_n \to X_i, (x_1,\ldots,x_n) \mapsto x_i$
- $\mathcal{P}(X) = \{U : U \subseteq X\}$ is the power set of X. A subset of $\mathcal{P}$ is a **system of subsets** of X.
- 'n-z' means 'non-zero'

Theorem: Random Theorems

- O-S: For G a group acting on a finite set X, $|\mathrm{Orb}(x)| = |G|/|\mathrm{Stab}(x)|$

Remark: Notation

- 'almost all' means 'all but finitely many'

Vector Spaces

Definition: Systems Of Linear Equations

Let F be a field. We seek to solve the system of n linear equations in m unknowns, described more easily by the extended coefficient matrix.

$$\begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m = b_2 \\ \vdots \qquad \vdots \qquad \vdots \qquad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m = b_n \end{array} \qquad \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1m} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2m} & b_2 \\ \vdots & & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} & b_n \end{array}\right)$$

- The set $L \subseteq F^m$ of m-tuples satisfying the equations is the **solution set**.
- If the solution set of a linsys of eqns $\neq \emptyset$ we obtain all solutions by adding componentwise an arbitrary solution of the homogenised system to a fixed solution of the system.

Definition: Matrices

A mapping $A : \{1,\ldots,n\} \times \{1,\ldots,m\} \to Z$ is called an $n \times m$-matrix with coefficients in Z.
- The set of all $n \times m$-matrices with coefficients in Z is denoted $\mathrm{Mat}(n \times m; Z) := \mathrm{Maps}(1,\ldots,n \times 1,\ldots,m,Z)$
- If $m = n$ we have a **square matrix** and write $\mathrm{Mat}(n,Z)$
- For square matrix A, its **trace** $\mathrm{tr}(A)$ is the sum of its diagonal entries
- A square matrix A is **nilpotent** with **index/degree** k if $A^k = 0$ for $k \in \mathbb{N}$.
- $P \in \mathrm{Mat}(n,\mathbb{R})$ is **orthogonal** if $P^T P = I_n$, ie $P^T = P^{-1}$.
- $P \in \mathrm{Mat}(n,\mathbb{C})$ is **unitary** if $\overline{P}^T P = I_n$, ie $\overline{P}^T = P^{-1}$.
- $P \in \mathrm{Mat}(n,\mathbb{C})$ is **hermitian** if $P = \overline{P}^T$, and the mapping on the standard IPS described by P is self-adjoint.
- For $A \in \mathrm{Mat}(n,\mathbb{C})$, $\exp(A) := \sum_{k=0}^{\infty} \frac{1}{k!} A^k$. For commutative A,B, $\exp(A+B) = \exp(A)\exp(B)$

Definition: Fields And Vector Spaces

A **field** F is a set with functions $+ : F \times F \to F; (\lambda,\mu) \mapsto \lambda + \mu$ and $\cdot : F \times F \to F; (\lambda,\mu) \mapsto \lambda\mu$ s.t. $(F,+)$ and $(F \backslash \{0\},\cdot)$ are abelian groups with respective identities $0_F, 1_F$, and $\lambda(\mu+\nu) = \lambda\mu + \lambda\nu \in F$.
- $0\vec{v} = \vec{0}; (-1)\vec{v} = -\vec{v}$
- $\forall \lambda \in F$, $\lambda\vec{0} = \vec{0}$
- If $\lambda\vec{v} = 0, \lambda = 0$ or $\vec{v} = \vec{0}$

A **vector space** V **over a field** F is a pair made up of an abelian group $V = (V,+)$ and a mapping $F \times V \to V; (\lambda,\vec{v}) \mapsto \lambda\vec{v}$ such that for all $\lambda,\mu \in F$ and $\vec{v},\vec{w} \in V$ the following hold:

$$\lambda(\vec{v}+\vec{w}) = (\lambda\vec{v}) + (\lambda\vec{w})$$
$$(\lambda+\mu)\vec{v} = (\lambda\vec{v}) + (\mu\vec{v})$$
$$\lambda(\mu\vec{v}) = (\lambda\mu)\vec{v}$$
$$1_F\vec{v} = \vec{v}$$

- Let X be a set and V an F-VS. Then the set $\mathrm{Maps}(X,V)$ of mappings $X \to V$ is an F-VS (define $(f+g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda(f(x))$).
- Given F-VSs $V_1,\ldots,V_n$, $V_1 \times \ldots \times V_n$ is an F-VS if we define addition/mult by scalars componentwise. We denote this VS by $V_1 \oplus \ldots \oplus V_n$ and name it the **external direct sum**. $F^n = F \oplus \ldots \oplus F$ n times.
- Two VsubSs V_1, V_2 of a VS V are **complementary** if addition defines a bijection: $V_1 \times V_2 \xrightarrow{\sim} V$

Definition: Vector Subspaces

A subset U of a vector space V is a **vector subspace** if $\vec{0} \in U$ and $\forall \vec{u},\vec{v} \in U, \lambda \in F$, $\vec{u}+\vec{v}$ and $\lambda\vec{u} \in U$.
- The solution set of a linsys of eqns seen earlier is a vector space
- An expression of form $\alpha_1\vec{v}_1 + \ldots + \alpha_n\vec{v}_n$ with n finite is a **linear combination** of $\vec{v}_1,\ldots,\vec{v}_n$.
- A subset is a **hyperplane** if it is a proper subspace and together with one other $\vec{v}$ generates the whole VS.
- The intersection of two VSs is itself a VS

Let T be a subset of V, a VS over F. $\langle T \rangle$, the **span of** T or **vector subspace generated by** T is the set of all linear combinations of vectors in T. (If $T = \emptyset, \langle T \rangle = \vec{0}$). Note if $v \in \langle T \rangle, \langle T \cup \{v\} \rangle = \langle T \rangle$.
A subset of V is a **generating/spanning set** if its span is V. A VS with a finite generating set is **finitely generated**.
- Given $V_1,\ldots,V_n$ VsubSs of V, we define the **sum** $V_1 + \ldots + V_n$ to be the subspace $\langle V_1 \cup \ldots \cup V_n \rangle$ of V. If the homo. given by addition: $V_1 \times \ldots \times V_n$ is also an injection the sum is **direct** and can be written $V_1 \oplus \ldots \oplus V_n$.

Definition: Linear Independence

A subset L of a vector space V is **linearly independent** if for all $\alpha_1,\ldots,\alpha_n \in F$ and distinct $\vec{v}_1,\ldots,\vec{v}_n \in L$, $\alpha_1\vec{v}_1 + \ldots + \alpha_n\vec{v}_n = 0$ only if all $\alpha_i = 0$. Otherwise it is **linearly dependent**.

Linearly Independent	Linearly Dependent
$\emptyset; \{\vec{v}\}, \vec{v} \neq \vec{0}$	$\{\vec{0}\}$

Definition: Bases

Let V be a VS over a field F. A **basis** of V is a linearly independent generating set in V.
- The family $(\vec{v}_i)_{1 \leq i \leq n}$ is a basis of V iff the 'evaluation mapping' $\Phi : F^n \to V; (\alpha_1,\ldots,\alpha_n) \mapsto (\alpha_1\vec{v}_1 + \ldots + \alpha_n\vec{v}_n)$ is a bijection.
- TFAE for a subset E of a VS V:
  - (i) E is a basis
  - (ii) E is minimal among generating sets ($\forall \vec{v} \in E$, $E \backslash \vec{v}$ does not generate V)
  - (iii) E is maximal among linearly independent subsets ($\forall \vec{v} \in E, E \cup \vec{v}$ is linearly dependent)
- If V is a finitely generated VS, it has a finite basis
- For a set X, the set $\mathrm{Maps}(X,F)$ equipped with pointwise addition and scalar multiplication is an F-VS. The subset, $F\langle X \rangle \subseteq \mathrm{Maps}(X,F)$, of all mappings which send almost all $x \in X$ to zero as a vector subspace, called the **free vector space on** X
- TFAE
  - (i) The family $(\vec{v}_i)_{i \in I}$ is a basis of V.
  - (ii) For all $v \in V$, $\exists$ precisely one family $(a_i)_{i \in I}$ s.t. almost all $a_i = 0$, and $\vec{v} = \sum_{i \in I} a_i\vec{v}_i$.

Theorem: FTLA, Steinitz, Exchange Lemma

Let V be a VS, $L \subset V$ linearly ind., and $E \subseteq V$ generating
- If V is finitely generated then it has a finite basis, and any two bases have the same cardinality
- Fundamental Theorem of Linear Algebra: $|L| \leq |E|$.
- Steinitz Exchange Theorem: There is an injection $\phi : L \hookrightarrow E$ s.t. $\{E \backslash \phi(L)\} \cup L$ is also generating.
- Exchange Lemma: Steinitz but for single elements.

Definition: Dimensions Of Vector Spaces

Let V be a finitely-generated VS, $L \subset V$ linearly ind., $E \subseteq V$ generating. For finitely-generated V, its **dimension**, $\dim V$ is the cardinality of a basis of V. For dimension as an F-VS write $\dim_F V$. Also:
- $|L| \leq \dim V$, if $|L| = \dim V$ then L is a basis
- $|E| \geq \dim V$, if $|E| = \dim V$ then E is a basis
- A proper vector subspace of V has a strictly smaller dimension than V
- For U,W subspaces of V, $\dim(U+W) + \dim(U \cap W) = \dim(U) + \dim(W)$.
- Given $V_1,\ldots,V_n$ F-VSs, $\dim(V_1 \oplus \ldots \oplus V_n) = \dim(V_1) + \ldots + \dim(V_n)$

MDfn: Linear Mappings: Homo/Iso/Endo/Automorphisms

Let V,W be VSs over a field F and $B \subset V$ be a basis. A mapping $f : V \to W$ is **(F-)linear** or a **homomorphism of** F**-vector spaces** if for all $\vec{v}_1,\vec{v}_2 \in V$ and $\lambda \in F$: $F(\vec{v}_1 + \vec{v}_2) = f(v_1) + f(v_2)$ and $f(\lambda\vec{v}_1) = \lambda f(\vec{v}_1)$. The set of all homomorphisms from V to W is $\mathrm{Hom}_F(V,W) = \mathrm{Hom}(V,W) \subseteq \mathrm{Maps}(V,W)$
- An **isomorphism** is a bijective linear mapping. Isomorphisms are invertible, and the inverse mapping is also an isomorphism
- An **endomorphism** is a homomorphism from a VS to itself
- An **automorphism** is an isomorphism from a VS to itself. The automorphisms of a VS V form a subgroup of its permutation group called the **general linear** or **automorphism** group, denoted $GL(V) = \mathrm{Aut}(V) \subseteq Maps^{\times}(V,V)$.
- If $f : V \to W$ and $g : U \to V$ are homo.s, $(f \circ g) : U \to W$ is a homo.
- The image and preimage of a VS under a linear mapping are VSs.
- A homo. can be restricted to a bijection $\mathrm{Hom}_F(V,W) \xrightarrow{\sim} \mathrm{Maps}(B,W)$, $f \mapsto f|_B$ - ie a linear mapping determines and is determined by the values it takes on a basis.
- Every injective homo. $f : V \hookrightarrow W$ has a **left inverse** g s.t. $g \circ f = \mathrm{id}_V$
- Every surjective homo. $f : V \twoheadrightarrow W$ has a **right inverse** g s.t. $f \circ g = \mathrm{id}_W$
- For $n \geq 0 \in \mathbb{Z}$, a VS V over F is isomorphic to F^n iff $\dim V = n$.
- A point sent to itself by a homo. is a **fixed point** of the mapping. Given a mapping $f : X \to X$, the set of fixed points is $X^f = \{x \in X : f(x) = x\}$.
- Given VS V, VsubSs $V_1,\ldots,V_n$, and linmaps $f_i : V_i \to V$; $f : V_1 \oplus \ldots \oplus V_n \to V$, $f(v1,\ldots,v_n) = f_1(v_1) + \ldots + f_n(v_n)$ is an isomorphism. We then write $V = V_1 \oplus V_2$ and say V is the **internal direct sum** of V_1, V_2.
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Definition: Image and Kernel

Let V,W be VSs and $f : V \to W$ be a linear mapping. The **image** of f is $(f) = F(V) \subseteq W$. It is a VsubS of W. The **kernel** of f is $\ker(f) := f^{-1}(\vec{0}) = \{\vec{v} \in V : f(\vec{v}) = \vec{0}\}$. It is a VsubS of V.
- A linear mapping f is injective iff its kernel is $\vec{0}$.
- The image of a VsubS under a linear mapping is again a VsubS, and the preimage of a VsubS under a linear mapping is again a VsubS.

Major Theorem: Rank-Nullity Theorem

Let V,W be VSs and $f : V \to W$ a linear mapping. Then $\dim V = \dim(\ker(f)) + \dim(\mathrm{im}(f))$. $\dim(\mathrm{im}(f))/\dim(\ker(f))$ are the **rank/nullity** of f.

Linear Mappings and Matrices

Theorem: Linking Linear Mappings And Matrices

Let F be a field and $m,n \in \mathbb{N}$. There is a bijection M between the space of linear mappings $F^m \to F^n$ and $\mathrm{Mat}(n \times m; F)$ which maps $f \mapsto [f]$. $M(f) := [f]$ is the **representing matrix** of f, whose columns are the images under f of the standard basis elements of F^m, ie $[f] = (f(\vec{e}_1)|\ldots|f(\vec{e}_m))$.

Theorem: Composing Mappings vs Matrix Multiplication

Let $g : F^\ell \to F^m$ and $f : F^m \to F^n$ be linmaps. The representing mat of their composition is the product of their representing mats: $[f \circ g] = [f][g]$.
- For $\ell,m,n \in \mathbb{N}$, F a field, $A \in \mathrm{Mat}(n \times m; F)$, $B \in \mathrm{Mat}(m \times \ell; F)$. Then $AB \in \mathrm{Mat}(n \times \ell)$ is defined by $(AB)_{ik} = \sum_{j=1}^{m} A_{ij}Bjk$.

Definition: The General Linear Group

The set of invertible $n \times n$ matrices in a field F form a group under matrix multiplication, called the **General Linear group of** $(n \times n)$ **matrices** and written $\mathrm{GL}(n;F) := \mathrm{Mat}(n;F)^{\times}$

Definition: Properties Of Matrices

- Let $k,\ell,m,n \in \mathbb{N}$, $A,A' \in \mathrm{Mat}(n \times m; F)$, $B,B' \in \mathrm{Mat}(m \times \ell; F)$, $C \in \mathrm{Mat}(\ell \times k; F)$, and $I = I_m$. Then:
  - (i) $(A+A')B = AB + A'B$
  - (ii) $A(B+B') = AB + AB'$
  - (iii) $(AB)C = A(BC)$
- A square matrix A is **invertible** if $\exists A^{-1}$ of same size s.t $AA^{-1} = A^{-1}A = I$
- An **elementary** matrix is one which differs from I in at most one entry
- Every square matrix with entries in a field can be written as a product of elementary matrices
- A matrix whose only non-zero entries lie on the diagonal, and which has first ones along the diagonal, then zeroes, is in **Smith Normal Form**. For each $A \in \mathrm{Mat}(n \times m; F)$, $\exists$ invertible P,Q s.t. PAQ is in Smith Normal Form.
- The **rank** of a matrix A is the dimension of the subspace of F^n generated by the columns or rows of A (the *dim* of each are always equal).
- A is of **full rank** if the rank is equal to the number of rows or columns
- $\mathrm{rk}(A+B) \leq \mathrm{rk}A + \mathrm{rk}B$, and $\mathrm{rk}(AB) \leq \min(\mathrm{rk}A, \mathrm{rk}B)$.

Definition: Representing Matrices W.R.T Bases

Let F be a field, V,W F-VSs with ordered bases $\mathcal{A} = (\vec{v}_1,\ldots,\vec{v}_m)$ and $\mathcal{B} = \vec{w}_1,\ldots,\vec{w}_n$. Then to each linmap $F : V \to W$ we associate a **representing matrix** ${}_{\mathcal{B}}[f]_{\mathcal{A}}$ with entries a_{ij} defined by $f(v_j) = a_{1j}\vec{w}_1 + \ldots + a_{nj}\vec{w}_n \in W$.
- This produces an isomorphism of VSs $M_{\mathcal{B}}^{\mathcal{A}} : \mathrm{Hom}_F(V,W) \xrightarrow{\sim} \mathrm{Mat}(n \times m; F)$ with $f \mapsto {}_{\mathcal{B}}[f]_{\mathcal{A}}$.
- We call $M_{\mathcal{B}}^{\mathcal{A}}(f) = {}_{\mathcal{B}}[f]_{\mathcal{A}}$ the **representing matrix of** f **with respect to the bases** $\mathcal{A}$ **and** $\mathcal{B}$. The columns of ${}_{\mathcal{B}}[f]_{\mathcal{A}}$ give the coords of the images of the basis vectors from $\mathcal{A}$ wrt the basis $\mathcal{B}$.
- Let U,V,W be finite-dimensional F-VSs with ordered bases $\mathcal{A},\mathcal{B},\mathcal{C}$, and $f : U \to V$, $g : V \to W$ be linmaps. Then the representing matrix of $(g \circ f) : U \to W$ is the matrix product of the representing matrices of f and g: ${}_{\mathcal{C}}[g \circ f]_{\mathcal{A}} = {}_{\mathcal{C}}[g]_{\mathcal{B}}{}_{\mathcal{B}}[f]_{\mathcal{A}}$
- Let $\mathcal{A}$ an ordered basis of a finite-dim VS V, the inverse of the 'evaluation mapping' $\Phi_{\mathcal{A}} : F^m \xrightarrow{\sim} V$, $(\alpha_1,\ldots,\alpha_m)^{\mathrm{T}} \mapsto \alpha_1\vec{v}_1 + \ldots + \alpha_m\vec{v}_m$ is given by $\vec{v} \mapsto {}_{\mathcal{A}}[\vec{v}]$. ${}_{\mathcal{A}}[\vec{v}]$ is the **representation of** $\vec{v}$ **w.r.t. the basis** $\mathcal{A}$.
- Let $\mathcal{A},\mathcal{B}$ be ordered bases of finite-dim VSs V,W and $f : V \to W$ be a linmap. Then for $\vec{v} \in V$, ${}_{\mathcal{B}}[f(\vec{v})] = {}_{\mathcal{B}}[f]_{\mathcal{A}}[\vec{v}]$

Definition: Change Of Matrix By Change Of Basis

Let $\mathcal{A} = (\vec{v}_1,\ldots,\vec{v}_n)$ and $\mathcal{B} = (\vec{w}_1,\ldots,\vec{w}_n)$ both be ordered bases of an F-VS V. Then the matrix representing the identity map w.r.t. these bases, ${}_{\mathcal{B}}[\mathrm{id}_V]_{\mathcal{A}}$ is called a **change of basis matrix**. By definition its entries are given by the equalities $\vec{v}_j = \sum_{i=1}^{n} a_{ij}\vec{w}_i$.
- Let V,W be finite-dim F-VSs and $f : V \to W$ be a linmap. Suppose $\mathcal{A},\mathcal{A}'$ and $\mathcal{B},\mathcal{B}'$ are ordered bases of V and W respectively. Then ${}_{\mathcal{B}'}[f]_{\mathcal{A}} = {}_{\mathcal{B}'}[\mathrm{id}_W]{}_{\mathcal{B}\mathcal{B}}[f]_{\mathcal{A}\mathcal{A}}[\mathrm{id}_V]_{\mathcal{A}'}$.
- For $\mathcal{A},\mathcal{A}'$ ordered bases of finite-dim VS V, and f an endo. of V; ${}_{\mathcal{A}'}[f]_{\mathcal{A}'} = {}_{\mathcal{A}}[f]_{\mathcal{A}'\mathcal{A}}^{-1}[f]_{\mathcal{A}\mathcal{A}}[\mathrm{id}_V]_{\mathcal{A}'}$
- $N = {}_{\mathcal{B}}[f]_{\mathcal{B}}$, $M = {}_{\mathcal{A}}[f]_{\mathcal{A}}$ are **similar** if $N = T^{-1}MT$ ($T := {}_{\mathcal{A}}[\mathrm{id}_V]_{\mathcal{B}}$).
- Let $f : V \to W$ be a linmap between finite-dim F-VSs. There exists ordered bases $\mathcal{A}$ of V and $\mathcal{B}$ of W s.t. ${}_{\mathcal{B}}[f]_{\mathcal{A}}$ is in Smith Normal Form

Rings And Modules

Major Definition: A Ring

A **ring** is a set R equipped with two operations $+,\cdot$ such that:
  - (i) $(R,+)$ is an abelian group
  - (ii) $(R,\cdot)$ is a monoid (satisfies associativity and identity)
  - (iii) $a \cdot (b+c) = (a \cdot b) + (a \cdot c)$ and $(a+b) \cdot c = (a \cdot c) + (b \cdot c)$.

The element $1 \in R$ is uniquely defined as the identity of $(R,\cdot)$.
- A ring in which $\cdot$ is commutative is a **commutative ring**.
- The **null/zero ring** is the ring where R is a single element set.
- The integers modulo m, or $\mathbb{Z}/m\mathbb{Z}$ is a ring, with congruence classes as elements.
- A field is a n-z commutative ring F where every n-z $a \in F$ has inverse $a^{-1} \in F$, ie $a \cdot a^{-1} = a^{-1} \cdot a = 1$
- For $m \in \mathbb{N}$, $\mathbb{Z}/m\mathbb{Z}$ is a field iff m is prime(
- For $a,b \in R$, $0a = 0 = a0$; $(-a)b = -(ab) = a(-b)$; and $(-a)(-b) = ab$.
- For $m \in \mathbb{Z}$, the m**-th multiple ma of** $a \in$ **abelian** R is $ma := a + \ldots + a$ m times if $m > 0$, $0a = 0$. If $m < 0$, $(-m)(a) = -(ma)$.
- For $m,n \in \mathbb{Z}$, $a,b \in R$: $m(a+b) = ma + mb$; $(m+n)a = ma + na$; $m(na) = (mn)a$; and $m(ab) = (ma)b = a(mb)$.

Major Definition: Units, Zero-Divisors, Integral Domains

Let R be a ring, and $a \in R$.
a is a **unit** if it is invertible in R ($\exists a^{-1} \in R$ s.t. $aa^{-1} = a^{-1}a = 1$).
A n-z a is a **zero-divisor** if $\exists$ n-z $b \in R$ s.t. either $ab = 0$ or $ba = 0$.
An **integral domain** is a n-z commutative ring with no zero-divisors.
- The set of units of a ring, $R^{\times}$, form a group under multiplication.
- For $m \in \mathbb{N}$, $\mathbb{Z}/m\mathbb{Z}$ is an integral domain iff m is prime
- Every finite integral domain is a field
- Let $a,b,c \in R$, an int-dom. If $ab = ac$ and $a \neq 0$, then $b = c$.

Definition: Polynomials

Let R be a ring. A **polynomial over** R is an expression of the form $P = a_m x^m + \ldots + a_2 x^2 + a_1 x + a_0$ for $0 \leq m \in \mathbb{Z}$ and $a_i \in R$.
- The set of all polynomials over R is denoted $R[x]$.
- When the leading coefficient is one, the polynomial is **monic**
- Defined with the usual addition/multiplication rules for polynomials, $R[x]$ becomes a ring, called the **ring of polynomials with coefficients in** R. The zero/identity of R are the zero/identity of $R[x]$.
- If R is commutative/an integral domain/has no zero-divisors, so is or does $R[x]$
- If R has no zero-divisors, $\deg(PQ) = \deg(P) + \deg(Q)$ for n-z $P,Q \in R[x]$.
- Division and Remainder Thm: Let R be an int-dom and $P,Q \in R[x]$ with Q monic. Then $\exists$ unique $A,B \in R[x]$ s.t. $P = AQ + B$; and $\deg(B) < \deg(Q)$ or $B = 0$
- For R a commutative ring $P \in R[x]$ can be **evaluated** at $\lambda \in R$ to give $P(\lambda)$. In this way we have a mapping $R[x] \to \mathrm{Maps}(R,R)$
- $\lambda \in R$ is a root of P if $P(\lambda) = 0$, and iff $(x - \lambda)$ divides $P(x)$.
- For R an int-dom, n-z $P \in R[x] \backslash \{0\}$ has at most $\deg(P)$ roots in R.

Definition: Algebraic Closure

A field F is **algebraically closed** if each non-constant polynom $P \in F[x] \backslash F$ with coefficients in F has a root in F.
- If F is algebraically closed, every n-z $P \in F[x] \backslash \{0\}$ decomposes uniquely (up to reordering) into linear factors: $P = c(x - \lambda_1)\ldots(x - \lambda_n)$ with $n \geq 0, c \in F^{\times}, \lambda_1,\ldots,\lambda_n \in F$.

Major Definition: Ring Homomorphisms

Let R,S be rings. A mapping $f : R \to S$ is a **ring homomorphism** if for all $x,y \in R$: $f(x+y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$.
- The **zero ring homomorphism** is $f : R \to S; r \mapsto 0$.
- **Isomorphisms** etc are akin to linear mappings
- If S has no zero-divs then either $f(1_R) = 0_S$ and f is the zero homo., or $f(1_R) = 1_S$.
- The composition of two homo.s is a homo., and the inverse of an iso. is an iso.
- If $f : R \to S$ a ring homo., then $\forall x,y \in R$ and $m \in \mathbb{Z}$: $f(0_R) = 0_S$; $f(-x) = -f(x)$; $f(x-y) = f(x) - f(y)$; $f(mx) = mf(x)$.

Major Definition: Ideals

A subset I of a ring R is an **ideal**, written $I \trianglelefteq R$, if it is nonempty; closed under subtraction; and closed under multiplication by elements of R.
- For R a comring and $T \subset R$, the **ideal of** R **generated by** T is the set ${}_R\langle T \rangle = \{r_1t_1 + \ldots + r_mt_m : t_1,\ldots,t_m \in T, r_1,\ldots,r_m \in R\}$ together with 0_R if $T = \emptyset$. If T is finite often write ${}_R\langle t_1,\ldots,t_n \rangle$.
- For R a comring and $T \subseteq R$, ${}_R\langle T \rangle$ is the smallest ideal of R containing T.
- For R a comring, an ideal I is a **principal ideal** if $I = \langle t \rangle$ for some $t \in R$. In a comring, $\{0\}$ and R are always principal ideals, generated by 0 and 1_R.

Definition: The Kernel Of A Ring Homomorphisms

Let R,S be rings with zero elements $0_R, 0_S$ and $f : R \to S$ be a ring homo. Then the **kernel** of f is $\ker f - \{r \in R : f(r) = 0_S\}$.
- $\ker f$ is an ideal of R.
- f is injective iff $\ker f = \{0\}$.
- The intersection of any collection of ideals is an ideal
- For I,J ideals of R; $I + J := \{a + b : a \in I, b \in J\}$ is an ideal

Definition: Subrings

Let R be a ring. A subset R' of R is a subring of R if R' is a ring under the same operations $+/\cdot$ defined in R.
Test For A Subring: A subset R' of a ring R is a subring if and only if:
  - (i) R' has a multiplicative identity ($1_{R'}$ is not necessarily equal to 1_R!)
  - (ii) R' is closed under subtraction ($a,b \in R' \Longrightarrow a - b \in R$)
  - (iii) R' is closed under multiplication

The intersections of two subrings of R is not necessarily a subring of R. Let R,S be rings and $f : R \to S$ a ring homomorphism. Then:
- If R' is a subring of R then $f(R')$ is a subring of S. Particularly, $\mathrm{im} f$ is a subring of S.
- Let $f(1_R) = 1_S$. Then if x is a unit in R, $f(x)$ is a unit in S and $(f(x))^{-1} = f(x^{-1})$. In this case f restricts to a group homomorphism $f|_{R^{\times}} : R^{\times} \to S^{\times}$.

Definition: Equivalence

A **relation** R on a set X is a subset $R \subseteq X \times X$. Only in this context, we write $(x,y) \in R$ as xRy. R is an **equivalence relation on** X when for all $x,y,z, \in X$:

(i) Reflexivity: xRx
(ii) Symmetry: xRy if and only if yRx
(iii) Transitivity: $(xRy$ and $yRz) \implies xRz$

* Let be an equivrel on a set X. For $x \in X$ the set $E(x) := \{z \in X : z\ x\}$ is the **equivalence class of** x. $E \subseteq X$ is an **equivalence class for** if $\exists x \in X$ s.t. $E = E(x)$.

- An element of an equivalence class is a **representative** of the class.
- $Z \subseteq X$ is a **system of representatives for** if it contains precisely one element from each equivalence class of .
- For $x,y \in X$ TFAE: $x\ y$; $E(x) = E(y)$; $E(x) \cap E(y) \neq \emptyset$

* Let be an equivrel on a set X. We denote the **set of equivalence classes of** X, a subset of $\mathcal{P}(X)$, as $(X/\ ) := \{E(x) : x \in X\}$.

- There is a surjective canonical mapping can: $X \twoheadrightarrow (X/\ ), x \mapsto E(x)$.
- Let be an equivrel on a set X. If $f : X \to Z$ is a mapping s.t. $x\ y \implies f(x) = f(y)$; there is a unique mapping $\overline{f} : (X/\ ) \to Z$, $\overline{f}(E(x)) := f(x)$, and $f = \overline{f} \circ$can.
- Often when defining a map $g : (X/\ ) \to Z$, we define $f : X \to Z$ and say g is **well-defined** to indicate that $x\ y \implies f(x) = f(y)$ as this guarantees the existence of $g = \overline{f}$.

Definition: Cosets and Factor Rings

Let $I \trianglelefteq R$ be an ideal in a ring R. The set $x + I := \{x + i : i \in I\} \subseteq R$ is a **coset of** I **in** R or the **coset of** X **with respect to** I **in** R.
Let R be a ring, $I \trianglelefteq R$ an ideal, and the equivrel defined by $x\ y \iff x - y \in I$. Then R/I, the **factor ring** or **quotient of** R **by** I, is the set $(R/\ )$ of cosets of I in R.
Let R be a ring and $I \trianglelefteq R$ an ideal. Then:

- R/I is a ring, with addition defined by $(x+I)+(y+I) = (x+y)+I$ and multiplication by $(x+I)(y+I) = xy+I$ for all $x,y \in R$.
- If R is commutative so is R/I
- R/I is a nonzero ring if and only if $I \neq R$
- If I is proper, and $r \in R^{\times}$; $r + I \in (R/I)^{\times}$ with $(r+I)^{-1} = r^{-1} + I$.

Theorem: The Universal Property Of Factor Rings

Let R be a ring and $I \trianglelefteq R$. Then:

1. The mapping can: $R \to R/I, r \mapsto r + I$ is a surjective ring homo. with kernel I.
2. If $f : R \to S$ is a ring homo. with $f(I) = \{0_S\}$, ie $I \subseteq \ker f$, there is a unique ring homo. $\overline{f} : R/I \to S$ s.t. $f = \overline{f} \circ$can.

$$\begin{tikzcd} R \arrow[r, "\text{can}"] \arrow[dr, "f"'] & R/I \arrow[d, dashed, "\overline{f}"] \\ & S \end{tikzcd}$$

Major Theorem: The First Isomorphism Theorem For Rings

Let R,S be rings. Then every ring homo. $f : R \to S$ induces a ring iso.: $\overline{f} : R/\ker f \xrightarrow{\sim} \operatorname{im} f$

Definition: Modules

A **(left) module** M **over a ring** R or an R**-module** is a pair consisting of an abelian group $M = (M, \dot{+})$ and a mapping $R \times M \to M; (r,a) \mapsto ra$ such that for all $r,s \in R$ and $a,b \in M$ the following hold:

- $r(a \dot{+} b) = (ra) \dot{+} (rb)$
- $(r+s)a = (ra) \dot{+} (sa)$
- $r(sa) = (rs)a$
- $1_R a = a$

Let R be a ring and M an $R-$module. Then for all $r \in R, a \in M$: $0_R a = 0_M$; $r 0_M = 0_M$; $(-r)a = r(-a) = -(ra)$

Definition: Module Homomorphisms/Kernel/Image/etc

Let R be a ring and M,N be $R-$modules.
A mapping $f : M \to N$ is an **(R-)homomorphism** if for all $a,b \in M$ and $r \in R$: $f(a+b) = f(a) + f(b)$, and $f(ra) = rf(a)$.
The **kernel** of f is $\ker f - \{a \in M : f(a) = 0_N\} \subseteq M$ and the **image** of f is $\operatorname{im} f = \{f(a) : a \in M\} \subseteq N$.
If f is a bijection then it is an **(R-module) isomorphism** with $M \cong N$.

- The composition of two homo.s is a homo.

Definition: Submodules

A non-empty subset M' of an $R-$module M is a **submodule** if M' is an $R-$module with respect to the operations of the $R-$module M restricted to M'.
Test For A Submodule:
Let R be a ring and M be an $R-$module. $M' \subseteq M$ is a submodule iff:

(i) $0_M \in M'$
(ii) $a,b \in M' \implies a - b \in M'$
(iii) $r \in R$ and $a \in M' \implies ra \in M'$

Let R be a ring, M,N be $R-$modules, $f : M \to N$ be an $R-$homo.. Then:

- $\ker f$ is a submodule of M and $\operatorname{im} f$ is a submodule of N.
- f is injective if and only if $\ker f = 0_M$

Let R be a ring, M an $R-$module, and $T \subseteq M$. Then the **submodule of** M **generated by** T is the set ${}_R\langle T \rangle = \{r_1 t_1 + \ldots + r_m t_m : t_1, \ldots, t_m \in T, r_1, \ldots, r_m \in R\}$ together with the zero element if $T = \emptyset$. Submodules have several properties akin to subspaces/ideals:

- M is **finitely generated** if it is generated by a finite set
- M is called **cyclic** if it is generated by a singleton
- ${}_R\langle T \rangle$ is the smallest submodule of M containing T, for $T \subseteq M$.
- The intersection of any collection of submodules of M is a submodule of M
- For M_1, M_2 submodules of M, $M_1 + M_2 := \{a + b : a \in M_1, b \in M_2\}$ is a submodule of M.
-

Definition: Factor Modules

Let R be a ring, M an $R-$module, and N a submodule of M. For each $a \in M$, the **coset of** a **with respect to** N **in** M is : $a + N = \{a + b : b \in N\}$. It is a coset of N in the abelian group M, so is an equiv class for the equivrel $a\ b \iff a - b \in N$. M/N, the **factor/quotient of** M **by** N to be the set $(M/\ )$ of all cosets of N in M. This becomes an $R-$module by defining: $(a+N) \dot{+} (b+N) := (a+b)+N$ and $r(a+N) = ra+N$. for all $a,b \in M, r \in R$.
The $R-$module M/N is the **factor module** of M by the submodule N.

Theorem: Universal Property Of Factor Modules

Let R be a ring, L,M be $R-$modules, and N a submodule of M:

1. The mapping can: $M \to M/N$, $a \mapsto a + N$ for all $a \in M$ is a surjective $R-$homo. with kernel N.
2. If $f : M \to L$ is an $R-$homo with $f(N) = \{0_L\}$ so that $N \subseteq \ker f$, there exists unique homo. $\overline{f} : M/N \to L$ such that $f = \overline{f} \circ$can.

$$\begin{tikzcd} M \arrow[r, "\text{can}"] \arrow[dr, "f"'] & M/N \arrow[d, dashed, "\overline{f}"] \\ & L \end{tikzcd}$$

Major Theorem: First Isomorphism Theorem For Modules

Let R be a ring and M,N be $R-$modules. Every $R-$homo. $f : M \to N$ induces an $R-$iso.: $\overline{f} : M/\ker f \xrightarrow{\sim} \operatorname{im} f$

- 2nd Theorem: For K,L sub-mods of M; L is a submod of $K + L = \{b + c : b \in K, c \in L\}$ and $K \cap L$ is a submod of K. Also, $\frac{K+L}{L} \cong \frac{K}{K \cap L}$
- 3rd Theorem: For K,L sub-mods of M, with $L \subseteq K$. Then K/L is a submod of M/L and $\frac{M/L}{K/L} \cong M/K$.

Determinants And Eigenvalues

Definition: Permutations And Their Signs

The group of all permutations of the set $\{1, \ldots, n\}$ to itself, denoted $\mathfrak{S}_n$, is the $n-$**th symmetric group**. It is a group under composition with order $n!$

- A **transposition** is a permutation which swaps two elements and leaves the others unchanged
- An **inversion** of a $\sigma \in \mathfrak{S}_n$ is a pair (i,j) s.t. $1 \leq i < j \leq n$ and $\sigma(i) > \sigma(j)$.
- The number of inversions of a σ is the **length** of σ, $\ell(\sigma)$
- The **sign** of σ is the parity of the number of inversions of σ - $\operatorname{sgn}(\sigma) = (-1)^{\ell(\sigma)}$.
- If $\operatorname{sgn}(\sigma) = 1$, σ is an **even permutation**. If it $= -1$, σ is an **odd permutation**
- $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$
- The set of even perms. in $\mathfrak{S}_n$ forms a subgroup of $\mathfrak{S}_n$ called the **alternating group** and denoted A_n.

Definition: Cofactors And Adjugate Matrices

Let $A \in \operatorname{Mat}(n;R)$ for a comring R, $n,i,j \in \mathbb{N}$ s.t. $1 \leq i,j \leq n$. Then:

- The (i,j) **cofactor** of A is $C_{ij} = (-1)^{i+j} \det(A\langle i,j \rangle)$ where $A\langle i,j \rangle$ is the matrix obtained by deleting the ith row and j^{th} column of A.
- The **adjugate matrix** $\operatorname{adj}(A)$ is the $(n \times n)$ matrix whose entries are $\operatorname{adj}(A)_{ij} = C_{ji}$ *(note order swap of i and j!).*

Major Definition: The Determinant

Let R be a comring and $n \in \mathbb{N}$. The **determinant** is a mapping $\det : \operatorname{Mat}(n;R) \to R$ given by: $\det(A) = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} \cdots a_{n\sigma(n)}$.
This is called the **Leibniz formula**. For $n = 0$ $\det(A) = 1$.
Alternative characterisations are:

- Let F be a field. The mapping $\det : \operatorname{Mat}(n;F) \to F$ is the unique alternating multilinear form on $n-$tuples of column vectors with values in F that takes the value 1_F on the identity matrix.
- Laplace Expansions: Let R be a comring and $A = (a_{ij}) \in \operatorname{Mat}(n;R)$. For fixed i,j the ith **row**/j^{th} **column expansion of the determinant** is $\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} / \sum_{i=1}^{n} a_{ij} C_{ij}$.

Theorem: Calculations With The Determinant

- Let R be a comring and $A,B \in \operatorname{Mat}(n;R)$. Then $\det(AB) = \det(A)\det(B)$; $\det(A^{\mathrm{T}}) = \det(A)$
- For A,B same-size square matrices with entries in a field F: $\det(A) \neq 0$ iff the A is invertible; if A is invertible then $\det(a^{-1}) = \det(A)^{-1}$ and $\det(A^{-1}BA) = \det(B)$ (ie, the det of an endomorphism is independent of the choice of basis used to calculate it)
- When performing Gaussian Elimination, row addition does not change the determinant, and row swaps change the sign only

Definition: Bi/Multilinear Forms

Let U,V,W be $F-$VSs. A **bilinear form on** $U \times V$ **with values in** W is a mapping $H : U \times V \to W$ which is linear in both entries, ie for all $u_1, u_2 \in U$ and $v_1, v_2 \in V$:

- $H(u_1 + u_2, v_1) = H(u_1, v_1) + H(u_2, v_1)$
- $H(u_1, v_1 + v_2) = H(u_1, v_1) + H(u_1, v_2)$
- $H(\lambda u_1, v_1) = \lambda H(u_1, v_1)$
- $H(u_1, \lambda v_1) = \lambda H(u_1, v_1)$

For $U = V$, a bilinear form is **symmetric** if for all $u,v \in U$, $H(u,v) = H(v_u)$; and it is **alternating/antisymmetric** if for all $u \in U$, $H(u,u) = 0$

- If H is antisymmetric, $H(u,v) = -H(v,u)$

Multilinear forms are just an extension of bilinear forms to higher dimensions - H must be linear in each argument.

- A multilinear form is **alternating** if $\exists i \neq j$ with $v_i = v_j \implies H(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_n) = 0$ and if so, has the property $H(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_n) = -H(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_n)$ for all $v_1, \ldots, v_n \in V$.
- Thus for any $\sigma \in \mathfrak{S}_n$, $H(v_{\sigma(1)}, \ldots, v_{\sigma(n)}) = \operatorname{sgn}(\sigma) H(v_1, \ldots, v_n)$

Theorem: Cramer's Rule

Let R a comring and $A \in \operatorname{Mat}(n;R)$. Then $A \cdot \operatorname{adj} A = (\det(A)) I_n$

- A is invertible iff $\det(A)$ is a unit in R, ie $\det(A) \in R^{\times}$

Theorem: Jacobi's Formula

Let $A = (a_{ij})$ where the coeffs $a_{ij} = a_{ij}(t)$ are functions of t. Then $\frac{d}{dt} \det(A) = \operatorname{tr} \operatorname{adj} A \frac{dA}{dt}$

Major Definition: Eigenvalues and Eigenvectors

Let $f : V \to V$ be an endomorphism of an $F-$VS V. $\lambda \in F$ is an **eigenvalue of** F iff there exists n-z $\vec{v} \in V$ s.t. $f(\vec{v}) = \lambda \vec{v}$. Each such $\vec{v} \in V$ is called an **eigenvector of** f **with eigenvalue** λ.
For any $\lambda \in F$, the **eigenspace of** f **with eigenvalue** λ is $E(\lambda, f) = \{\vec{v} \in V : f(\vec{v}) = \lambda \vec{v}\}$.

- Each endomorphism of a n-z finite-dim VS over an algebraically closed field has an eigenvalue
- If $\vec{v}_1, \ldots, \vec{v}_n$ are eigenvecs of f with distinct eigenvals $\lambda_1, \ldots, \lambda_n$, then $\vec{v}_1, \ldots, \vec{v}_n$ are linearly independent.

Definition: The Characteristic Polynomial

Let R be a comring and $A \in \operatorname{Mat}(n;R)$. The polynomial $\chi_A(x) := \det(xI_n - A) \in R[x]$ is called the **characteristic polynomial of** A.

- For F a field and $A \in \operatorname{Mat}(n;F)$. The eigenvalues of the linmap $A : F^n \to F^n$ are exactly the roots of χ_A.

Definition: Conjugacy

For a comring R, $A,B \in \operatorname{Mat}(n;R)$, are **conjugate** if $B = P^{-1}AP$ for invertible $P \in \operatorname{GL}(n;R)$.

- Conjugacy is an equivalence relation on $\operatorname{Mat}(n;R)$
- The characteristic polynomials of conjugate matrices are the same

Definition: Triangularisability

Let $f : V \to V$ be an endomorphism of a finite-dim $F-$VS V. The following two statements are equivalent, and if they hold, f is **triangularisable**:

(1) The characteristic polynom of f, $\chi_f(x)$, decomposes into linear factors in $F[x]$.
(2) V has an ordered basis $\mathcal{B} = (\vec{v}_1, \ldots, \vec{v}_n)$ s.t. $f(\vec{v}_1) = a_{11}\vec{v}_1$, $f(\vec{v}_2) = a_{12}\vec{v}_1 + a_{22}\vec{v}_2, \ldots, f(\vec{v}_n) = a_{1n}\vec{v}_1 + \ldots + a_{nn}\vec{v}_n \in V$. Equivalently, the matrix ${}_{\mathcal{B}}[f]_{\mathcal{B}}$ is upper triangular.

- An endomorphism $A : F^n \to F^n$ is triangularisable iff A is conjugate to an upper triangular matrix B. Thus any endomorphism of a finite-dim VS over an algebraically closed field is triangularisable.

Definition: Diagonalisability

An endo. $f : V \to V$ of an $F-$VS is **diagonalisable** iff there exists a basis of V consisting of eigenvectors of f. If V is finite-dim, then this is equivalent to: $\exists$ ordered basis $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_n\}$ such that ${}_{\mathcal{B}}[f]_{\mathcal{B}} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Here $f(\vec{v}_i) = \lambda_i \vec{v}_i$.
A square matrix $A \in \operatorname{Mat}(n;F)$ is **diagonalisable** iff the linear mapping $F^n \to F^n$ given by left multiplication by A is diagonalisable. That is, A is conjugate to a diagonal matrix ($\exists P \in \operatorname{GL}(n;F)$ s.t. $P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.) In this case the columns of P are the vecs of a basis of F^n consisting of eigenvecs of A with eigenvals $\lambda_1, \ldots, \lambda_n$.

Theorem: Cayley-Hamilton Theorem

Let R be a comring and $A \in \operatorname{Mat}(n;R)$. Then evaluating $\chi_A(x)$ at the matrix A gives zero.

Inner Product Spaces

Major Definition: Inner Products And Their Spaces

Let V be an $\mathbb{R}-$VS. An **inner product** on V is a mapping $(-,-) : V \times V \to \mathbb{R}$ that satisfies, for all $\vec{x}, \vec{y}, \vec{z} \in V$ and $\lambda, \mu \in \mathbb{R}$:
(1) $(\lambda\vec{x} + \mu\vec{y}, \vec{z}) = \lambda(\vec{x}, \vec{z}) + \mu(\vec{y}, \vec{z})$
(2) $(\vec{x}, \vec{y}) = (\vec{y}, \vec{x})$
(3) $(\vec{x}, \vec{x}) \geq 0$, $= 0$ iff $\vec{x} = 0$

- The **standard inner product** on $V = \mathbb{R}^n$ is the dot product
- A **real inner product space** is a real vector space endowed with an inner product
- $(-,-)$ is a symmetric bilinear form (1),(2) and **positive definite** (3).
- For V a VS over $\mathbb{C}$ the definition is similar, swap $\mathbb{R}$ for $\mathbb{C}$ everywhere; condition (2) reads $(\vec{x}, \vec{y}) = \overline{(\vec{y}, \vec{x})}$; and the standard inner product is $(\vec{v}, \vec{w}) = v_1\overline{w}_1 + \ldots + v_n\overline{w}_n$. Note the SIP is not $\mathbb{C}-$linear in the second variable.
- In an $\mathbb{R}$ or $\mathbb{C}-$VS, we define the **length/inner product norm/norm** as $\|\vec{v}\| := \sqrt{(\vec{v}, \vec{v})}$
- Vectors of length 1 are **units**
- $\vec{v}, \vec{w}$ are **orthogonal**, $\vec{v} \perp \vec{w}$, iff $(\vec{v}, \vec{w}) = 0$. Sets S,T are orthogonal, $S \perp T$, if $\vec{v} \perp \vec{w} \forall \vec{v} \in S, \vec{w} \in T$.
- A family $(\vec{v}_i)_{i \in I}$ for vecs from an $IP-$space is an **orthonormal family** if all $\vec{v}_i$ have length 1 and are pairwise orthogonal (using the Kronecker delta, $(\vec{v}_i, \vec{v}_j) = \delta_{ij}$
- An orthonormal family that is a basis is an **orthonormal basis**. For V an IPS and $(\vec{v}_i)_{i \in I}$ an orthonormal basis, for any $\vec{w} \in V$, $\vec{w} = \sum_{i \in I} (\vec{w}, \vec{v}_i) \vec{v}_i$.
- Every finite-dim inner product space has an orthonormal basis

Dfn: Orthogonal Complements/Projections

Let V be an IPS and $T \subseteq V$. Define $T^{\perp} = \{\vec{v} \in V : \vec{v} \perp \vec{t} \forall \vec{t} \in T\}$, the **orthogonal** to T. $T^{\perp}$ is always a subspace of V even if T is not, and $T^{\perp} = \langle T \rangle^{\perp}$.
Let U be a finite-dim subspace of an IPS V. The space $U^{\perp}$ is the **orthogonal complement to** U.

- Every $\vec{v} \in V$ can be written as $\vec{v} = \vec{p} + \vec{r}, \vec{p} \in U$ and $\vec{r} \in U^{\perp}$.
- For V an IPS and U a finite-dim subspace of V, U and $U^{\perp}$ are complementary, ie $V = U \oplus U^{\perp}$.

The **orthogonal projection from** V **onto** U is the mapping $\pi_U : V \to V, \vec{v} = \vec{p} + \vec{r} \mapsto \vec{p}$.

- π_U is a linmap with $\operatorname{im}(\pi_U) = U$ and $\ker(\pi_U) = U^{\perp}$
- $\pi_U^2 = \pi_U$, ie π_U is idempotent
- If $\{\vec{v}_1, \ldots, \vec{v}_n\}$ is an orthonormal basis of U, then for all $\vec{v} \in V$, $\pi_U(\vec{v}) = \sum_{i=1}^{n} (\vec{v}, \vec{v}_i) \vec{v}_i$.

Major Theorem: The Cauchy-Schwarz Inequality

Let $\vec{v}, \vec{w}$ be vectors in an inner product space.
Then $|(\vec{v}, \vec{w})| \leq \|\vec{v}\| \|\vec{w}\|$ with equality iff $\vec{v}, \vec{w}$ are linearly dependent.
For V an IPS, $\vec{v}, \vec{w} \in V$, and λ a scalar (ie in the underpinning field):

- $\|\vec{v}\| \geq 0$, $= 0$ iff $\vec{v} = \vec{0}$.
- $\|\lambda\vec{v}\| = |\lambda| \|\vec{v}\|$
- $\|\vec{v} + \vec{w}\| \leq \|\vec{v}\| + \|\vec{w}\|$.
- In a real IPS, $\exists$ unique $0 \leq \theta \leq \pi$ s.t. $\cos(\theta) = \frac{(\vec{v}, \vec{w})}{\|\vec{v}\| \|\vec{w}\|}$.

Theorem: Gram-Schmidt Process

Let $\vec{v}_1, \ldots, \vec{v}_k$ be linearly ind. vectors in an IPS V. Then we can construct an orthonormal family $\vec{w}_1, \ldots, \vec{w}_k$ such that for all $1 \leq i \leq k$, $\langle \vec{v}_1, \ldots, \vec{v}_i \rangle = \langle \vec{w}_1, \ldots, \vec{w}_i \rangle$. If V is finite-dim, we can use the G-S process to transform any basis into an orthonormal basis.

- For $k = 1$ take $\vec{w}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|}$ (normalise the length)
- Then given $\vec{w}_1, \ldots, \vec{w}_k$; $\vec{w}_{k+1} = \frac{\vec{v}_{k+1} - \sum_{i=1}^{k} (\vec{v}_{k+1}, \vec{w}_i) \vec{w}_i}{\left\| \vec{v}_{k+1} - \sum_{i=1}^{k} (\vec{v}_{k+1}, \vec{w}_i) \vec{w}_i \right\|}$

Definition: Adjoints and Self-Adjointness

Let V be an IPS. Two endomorphisms $T,S : V \to V$ are **adjoint** if for all $\vec{v}, \vec{w} \in V$, $(T\vec{v}, \vec{w}) = (\vec{v}, S\vec{w})$. In this case $S = T*$ and S is the **adjoint of** T.

- Let V be a finite-dim IPS and $T : V \to V$ an endo. Then $T*$ exists an is unique.
- An endo. of an IPS $T : V \to V$ is **self-adjoint** if $T = T*$.
- Let $T : V \to V$ be an self-adjoint linmap on an IPS V. Then:
  (1) Every eigenvalue of T is real
  (2) If λ, μ are distinct eigvals of T with eigenvecs $\vec{v}, \vec{w}$, then $(\vec{v}, \vec{w}) = 0$
  (3) T has an eigenvalue.

Major Theorem: The Spectral Theorem

Let V be a finite-dim IPS and $T : V \to V$ be a self-adjoint linmap. Then V has an orthonormal basis consisting of eigenvectors of T.

- Let $A \in \operatorname{Mat}(n, \mathbb{R})$ be symmetric. Then $\exists$ orthogonal $P \in \operatorname{Mat}(n, \mathbb{R})$ such that $P^T AP = P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, where the λ_i are the (necessarily real) eigvals of A repeated according to their multiplicity as roots of χ_A.
- Let $A \in \operatorname{Mat}(n, \mathbb{C})$ be hermitian. Then $\exists$ unitary $P \in \operatorname{Mat}(n, \mathbb{C})$ s.t. $\overline{P}^T AP = P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, where the λ_i are the (necessarily real) eigvals of A repeated according to their multiplicity as roots of χ_A.

Jordan Normal Form

In this section assume that F is an algebraically closed field

Definition: Jordan Blocks

Let $r \in \mathbb{N}$. Then

- The **nilpotent Jordan block of size** r is the $(r \times r)$ matrix $J(r)$ defined by the rule $J(r)_{ij} = 1$ for $j = i + i$ and $= 0$ otherwise
- The **Jordan block of size** r **and eigenvalue** λ, for $\lambda \in F$, is the $(r \times r)$ matrix $J(r, \lambda)$ defined by the rule $J(r, \lambda) = \lambda I_r + J(r) = D + N$ (with $DN = ND$).

Major Theorem: Jordan Normal Form

Let F be an algebraically closed field, V be a finite-dim VS, and $\phi : V \to V$ an endomorphism with characteristic polynomial $\chi_\phi(x) = (x - \lambda_1)^{a_1}(x - \lambda_2)^{a_2} \ldots (x - \lambda_s)^{a_s} \in F[x]$, $a_i \geq 1, \sum_{i=1}^{s} = n$ for distinct $\lambda_1, \ldots, \lambda_s \in F$. Then there
exists an ordered basis $\mathcal{B}$ of V such that the matrix of ϕ with respect to the basis $\mathcal{B}$ is block diagonal with Jordan blocks on the diagonal. That is, ${}_{\mathcal{B}}[\phi]_{\mathcal{B}} = \mathrm{diag}(J(r_{11}, \lambda_1), \ldots, J(r_{1m_1}, \lambda_1), J(r_{21}, \lambda_2), \ldots, J(r_{sm_s}, \lambda_s))$ with $r_{11}, \ldots, r_{1m_1}, r_{21}, \ldots, r_{sm_s} \geq 1$
such that $a_i = r_{i1} + r_{i2} + \ldots + r_{im_i}, (1 \leq i \leq s)$.

Remark: Overview of the Proof of Jordan Normal Form

he rough steps of the proof of the Jordan Normal Form are as follows:

(1) Decompose V into a direct sum $V = \oplus_{i=1}^{s} V_i$ according to the factorisation of χ_ϕ as a product of linear factors, where for each i:
   - $V_i = \ker((\phi - \lambda_i \mathrm{id}_V)^{a_i} : V \to V) \subseteq V$
   - $\phi(V_i) \subseteq V_i$
   - $(\phi - \lambda_i \mathrm{id}_{V_i})^{m_i}$ is zero on V_i for large enough m_i.

(2) We can now focus attention on the individual spaces v_i. These spaces have the advantage that a power of the endomorphism $(\phi - \lambda_i \mathrm{id}_{V_i}) : V_i \to V_i$ is zero. Ie, $\psi := \phi - \lambda_i \mathrm{id}_{V_i}$ is a nilpotent linmap on V_i.

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